

ADITYA KANE

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AREAS OF INTEREST

Efficiency in Deep Learning, Hardware-aware Co-design of Efficiency Techniques

EDUCATION

Georgia Institute of Technology

Ph.D. in Computer Science

Advisor: Prof. Humphrey Shi

Incoming Fall 2025

Georgia Institute of Technology

MS in Computer Science | Thesis under Prof. Humphrey Shi on fused kernels for Rotary Embeddings.

Coursework: ML with Limited Supervision, Efficient ML, Explainable AI

GPA: **4.0 / 4.0** | 2023 - 2025

Pune Institute of Computer Technology, Pune

Bachelor of Engineering in Computer Engineering with Honors in Artificial Intelligence and Machine Learning

CGPA: **9.7 / 10** | 2019-2023

EXPERIENCE

Intern, Member of Technical Staff | Yutori, San Francisco

May 2025 - August 2025

Dhruv Batra | Distributed training and inference for web agents

- Worked on implementing and verifying a FSDP2 + sequence-parallel based training framework for long context VLM training.
- Achieved strong throughput scaling up to 64 H100 GPUs, 1.4× to 2× E2E training speedup compared to DeepSpeed's FSDP1.
- Worked on setting up, stress-testing and modeling latency of hosted vLLM inference endpoints for production deployment.

Graduate Student Researcher | SHI Labs, Atlanta

August 2023 - Present

Prof. Humphrey Shi | High Performance AI and Computer Vision

- Working on **efficiency of large attention-based models**. Exploring model and hardware co-design of efficiency techniques.
- Contributed to *NATTEN*, an open-source library for fast implementations of static locality-based sparse attention.
- Developed fused CUDA kernels to speed up Rotary Position Embeddings (RoPE) and incorporated them to vision transformers.
- Head Graduate Teaching Assistant for Introduction to Computer Vision from Fall 2023 to Spring 2025.

Graduate Research Assistant | SHI Labs, Atlanta

July 2024 - Present

Prof. Humphrey Shi | Computer Vision

- Working on improving speed and accuracy of detection and tracking system for joint body and face tracking.
- Improved system throughput by > 5× with no loss in accuracy, and by > 20× with marginal loss in accuracy through minimizing redundant slow-path computations, maximizing on-device computations and interpolation of frames.

Research Intern | Indian Institute of Science, Bengaluru

May 2022 - December 2022

Prof. Suresh Sundaram and Dr. Chandan Gautam | Out-of-Distribution (OOD) Detection and Open Set Recognition in NLP

- Worked on **Out-of-Distribution and Open Set detection** in NLP at Artificial Intelligence and Robotics Lab, IISc, Bengaluru.
- Developed and published state-of-the-art out of scope intent detection methods, beating prior methods by significant margins.

Student Developer Intern at TensorFlow, KerasCV | Google Summer of Code

Summer 2021, 2022

Sayak Paul, Morgan Roff, Luke Wood | Computer Vision

- Mentee for Google Summer of Code 2021 and 2022 at the TensorFlow and Keras organizations.
- Implemented and trained four variants of RegnetY from the paper "**Designing network design spaces**" on **ImageNet-1k** using **TensorFlow 2**. Later ported over to Keras and KerasCV. Project reports available [here](#) and [here](#).
- Implemented various model and preprocessing pipeline components and merged them to KerasCV. Added a new **robust serialization test** which eliminated the need of hand-engineered tests for new modules.

Research Assistant | PICT, Pune

October 2020 - July 2023

Prof. Geetanjali Kale | Object detection

- Working on the research project "Question Wise segmentation of Handwritten examination paper in AI-Assisted Grading System".
- Developed a system and dataset for text localization in handwritten documents. Outperformed off-the-shelf object and text detectors.

PUBLICATIONS & RESEARCH

[C: Conference, W: Workshop]

C1. Generalized Neighborhood Attention: Multi-dimensional Sparse Attention at the Speed of Light

- Under review

C2. Revisiting Positional Information in Vision Transformers in the Era of Fused Attention

· Under review

C3. Class Name Guided Out-of-Scope Intent Classification

[Findings] EMNLP 2024

· Authors: Chandan Gautam, Sethupathy Parameswaran, **Aditya Kane**, Yuan Fang, Savitha Ramasamy, Suresh Sundaram, Sunil Kumar Sahu, Xiaoli Li

[Paper](#)

C4. Unsupervised Out-of-Distribution Detection Using Few In-Distribution Samples

[Poster] ICASSP 2023

· Authors: Chandan Gautam, **Aditya Kane**, Savitha Ramasamy, Suresh Sundaram

[Paper](#)

C5. My Boli: Code-mixed Marathi-English Corpora, Pretrained Language Models and Evaluation Benchmarks

[Findings] IJCNLP-AACL 2023

· Authors: Tanmay Chavan*, Omkar Gokhale*, **Aditya Kane***, Shantanu Patankar, Raviraj Joshi

[Paper](#)

W1. Task Arithmetic with LoRA for Continual Learning

[Poster] WANT, R0-FoMo workshops NeurIPS 2023

· Authors: Rajas Chitale*, Ankit Vaidya*, **Aditya Kane***, Archana Ghotkar

October 2023

[Paper](#) | [Code](#)

W2. Spread Love Not Hate: Undermining the Importance of Hateful Pre-training for Hate Speech Detection

[Poster] I Can't Believe It's Not Better Workshop NeurIPS 2022

· Authors: Omkar Gokhale*, **Aditya Kane***, Shantanu Patankar*, Tanmay Chavan*, Raviraj Joshi

[Paper](#)

W3. Continual VQA for Disaster Response Systems

[Poster] Tackling Climate Change with Machine Learning Workshop NeurIPS 2022

· Authors: **Aditya Kane***, V Manushree*, Sahil Khose*

October 2022

[Paper](#) | [Code](#) | [WandB](#)

PROJECTS

Added RegNets to `tf.keras.applications`

January 2022

Computer Vision | Collaborators: [Sayak Paul](#)

- Extended my project from Google Summer of Code. Trained **24** variants of RegNets on the **ImageNet-1k** dataset.
- These models are now added to `tf.keras.applications` and are available in TensorFlow 2.9 and later.

Course projects at Georgia Tech

December 2023

Computer Vision

1. **Machine Unlearning**: Course project for [Machine Learning with Limited Supervision](#) (Instructor: Prof. Judy Hoffman). Worked on removing the influence of a few samples from a trained model, or to make the model "unlearn" those datapoints. Project video available [here](#).
2. **OOD detection using model explanations in ViTs**: Course project for Explainable AI (Instructor: Prof. Sonia Chernova). We use attention rollout and gradients of attention maps to determine whether a given input is out-of-distribution.
3. **Using fewer visual tokens in LLaVA**: Course project for [Efficient Machine Learning](#) (Instructor: Prof. Celine Lin). We reduce image tokens in LLaVA using Token Merging. We retain accuracy and improve model performance without any retraining. Project poster [here](#).

VOLUNTEER EXPERIENCE

TensorFlow User Group (TFUG) Pune

September 2021 - July 2023

- Co-organizer of TensorFlow User Group, Pune. TFUGs are local communities of students and practitioners of the TensorFlow library. Our community has **3000+ registered members**. We enjoy a turnout of around 40 people for each monthly event.
- Received the **"Most Impactful Community Leader"** award by Google for consistently organizing most TFUG events in 2021.

Reviewing

- **Reviewer for ICML 2024 Workshop on Advancing Neural Network Training**: Reviewed three papers for the [WANT @ ICML 2024](#).
- **Book review**: Reviewed the book ["Practical Deep Learning for Computer Vision with Python"](#) by [David Landup](#).

TECHNICAL SKILLS

Languages	Python, C++, CUDA, PHP, SQL, HTML
Frameworks	NumPy, TensorFlow, Keras, PyTorch, JAX, Flax
Others	Git and GitHub, Google Cloud